Number Of Binary Tree Topologies

AlgoExpert · Very Hard · Recursion

The problem

- Number of structurally distinct binary trees with n nodes (Catalan).

Intuition

- Pick how many nodes go left; the counts multiply, giving the Catalan numbers.

Approach

- 1. ways of zero is one
- 2. For each n, sum left times right counts
- 3. Return ways of n

Java solution

```java
public class NumberOfBinaryTreeTopologies {
    public static long numberOfBinaryTreeTopologies(int n) {
        long[] c = new long[n + 1];
        c[0] = 1;
        for (int m = 1; m <= n; m++) for (int l = 0; l < m; l++) c[m] += c[l] * c[m - 1 - l];
        return c[n];
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n)

Recap

- Pick how many nodes go left; the counts multiply, giving the Catalan numbers.
- Complexity: O(n^2)